

AccFG: Accurate Functional Group Extraction and Molecular Structure Comparison

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Introduction – What is functional group

- Functional groups (FGs) are specific groups of atoms within molecules that impart unique physical and chemical properties
 - hydroxyl groups (-OH) are known for their polarity and ability to form hydrogen bonds,
 - carboxylic groups (-COOH) play a critical role in numerous organic reactions.
 - $\text{-OH} + \text{-COOH} \Rightarrow \text{Esterification} \text{-COO-}$
- Why FGs?
 - Language side
 - Prior knowledge in text format
 - Molecule side
 - Uncover structure activity relationship

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Hydroxy group

61 languages

Article Talk

From Wikipedia, the free encyclopedia

This article is about the hydroxy functional group. For the Hydroxyl radical, see Hydroxyl radical.

In **chemistry**, a **hydroxy** or **hydroxyl group** is a functional group with the chemical formula -OH and composed of one oxygen atom covalently bonded to one hydrogen atom. In organic chemistry, alcohols and carboxylic acids contain one or more hydroxy groups. Both the negatively charged anion HO^- , called **hydroxide**, and the neutral radical $\text{HO}^\cdot$, known as the **hydroxyl radical**, consist of an unbonded hydroxy group.

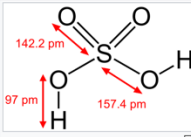
According to IUPAC definitions, the term *hydroxyl* refers to the hydroxyl radical (-OH) only, while the functional group -OH is called a *hydroxy group*.^[1]

Properties

Water, alcohols, carboxylic acids, and many other hydroxy-containing compounds can be readily deprotonated due to a large difference between the electronegativity of oxygen (3.5) and that of hydrogen (2.1). Hydroxy-containing compounds engage in intermolecular hydrogen bonding increasing the electrostatic attraction between molecules and thus to higher boiling and melting points than found for compounds that lack this functional group. Organic compounds, which are often poorly soluble in water, become water-soluble when they contain two or more hydroxy groups, as illustrated by sugars and amino acid.^[citation needed]

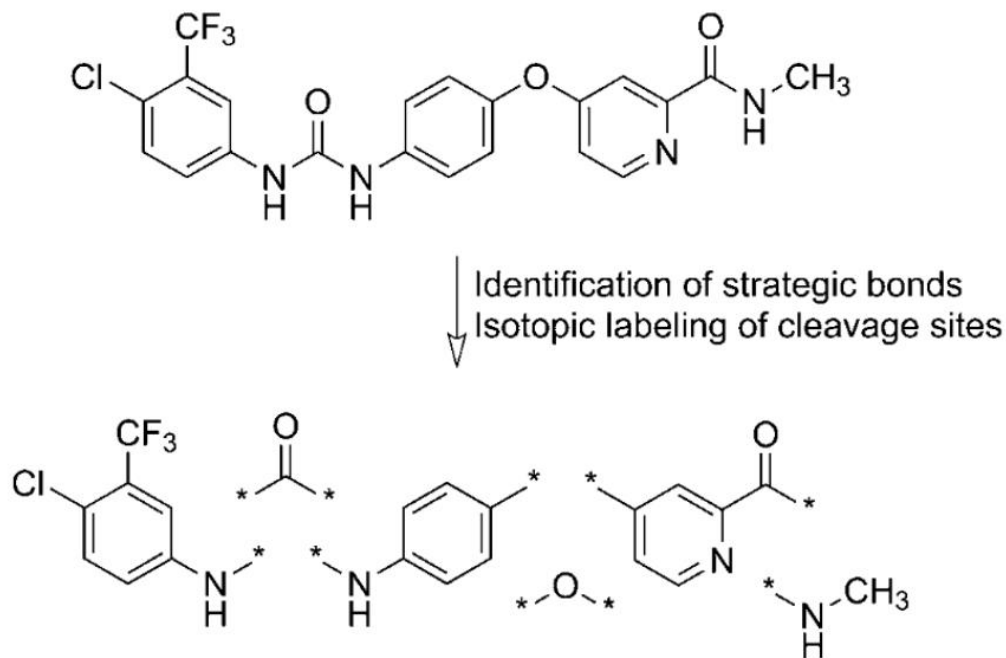
Representation of an organic compound hydroxy group, where R represents a hydrocarbon or other organic moiety, the red and grey spheres represent oxygen and hydrogen atoms respectively, and the rod-like connections between these, covalent chemical bonds.

Sulfuric acid contains two hydroxy groups.

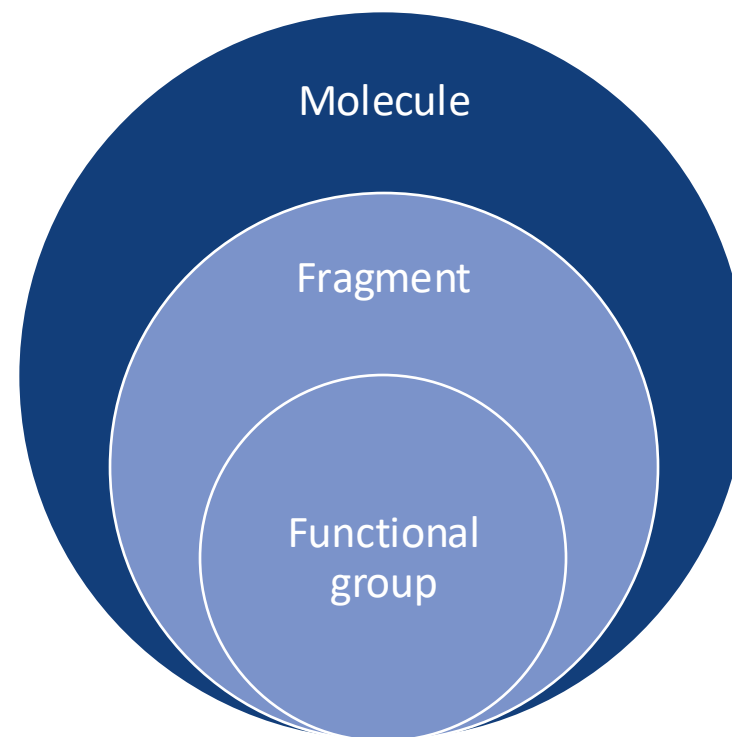


Fragments vs. Functional groups

BRICS method for molecule fragmentation



Ambiguity and Dynamic Nature:
Functional groups are “well-studied” fragments



Fine-grained molecular annotation enables interpretable connections between structure and chemical knowledge

- **Limitations of traditional models:**

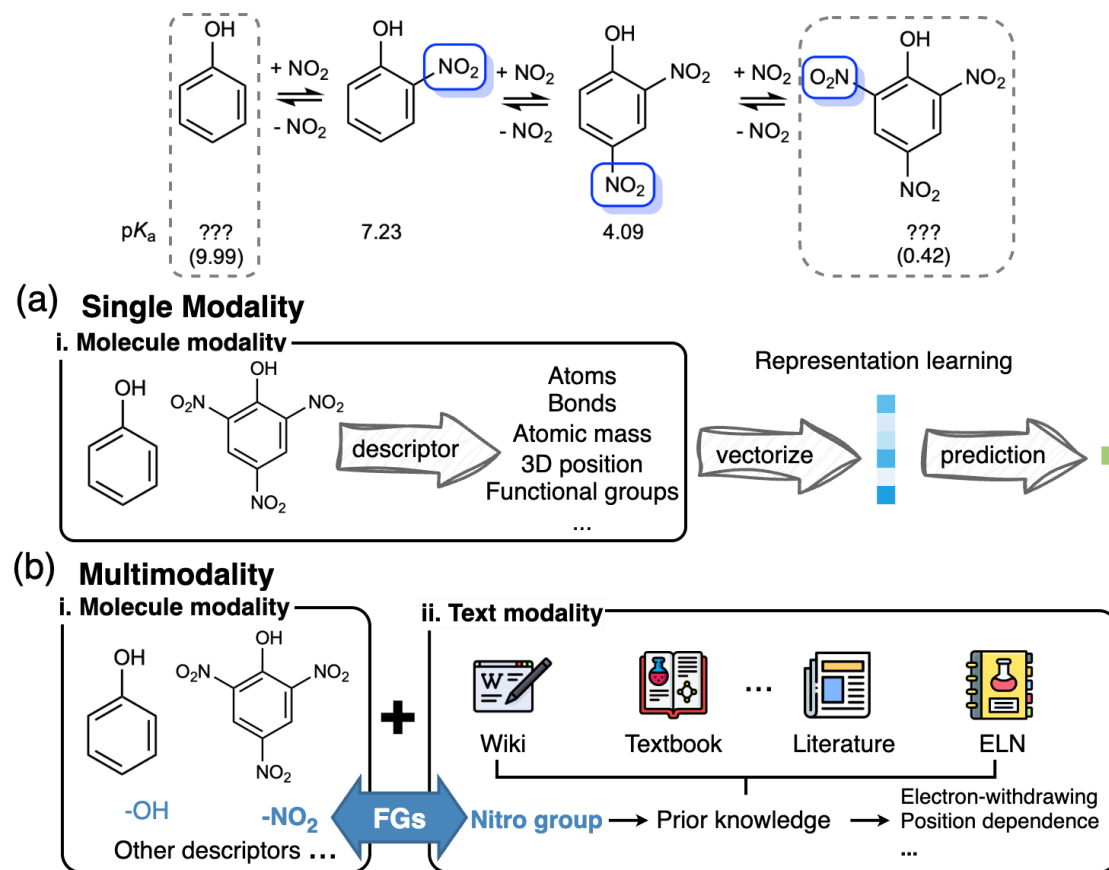
- Rely solely on molecular modality.
- Lack interpretability for humans and text-based models.
- Cannot integrate established domain knowledge.

- **Functional groups as a bridge:**

- Encode interpretable chemical features (e.g., electron-withdrawing $-\text{NO}_2$).
- Connect molecular structures to textual information in wikis, textbooks, and lab notes.

- **Outcome:**

- Enables multimodal reasoning combining structure + text.
- Supports interpretable and chemically grounded predictions.
- Establishes functional group identification as a foundation for multimodal chemical understanding.



An Accurate Molecule Parser Is Required.

Key Capabilities of AccFG

Overview

• AccFG enables:

- Accurate functional group extraction
- Functional group-level molecular comparison

Key Innovations

• Tailored algorithms:

- Atom & bond mapping
- Synonym merging
- Hierarchy construction

• Addresses redundancy and improves readability over prior methods

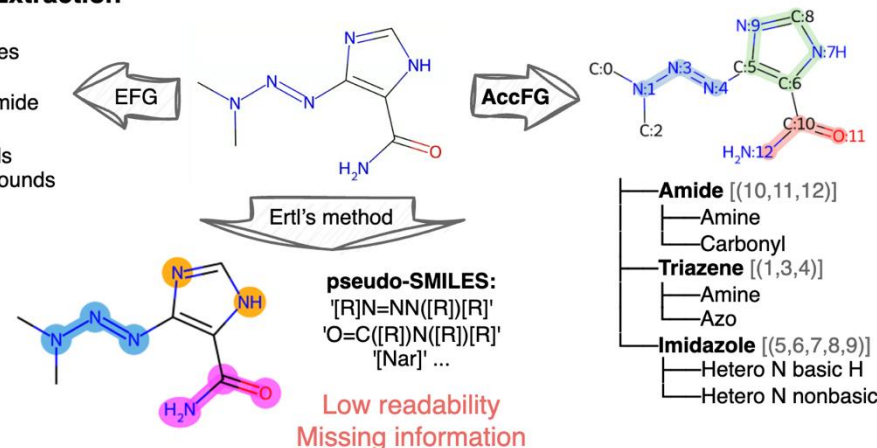
• Infrastructure-Level Utility

- Bridges molecular modality with domain knowledge in textual modality
- Designed for broad applicability in cheminformatics

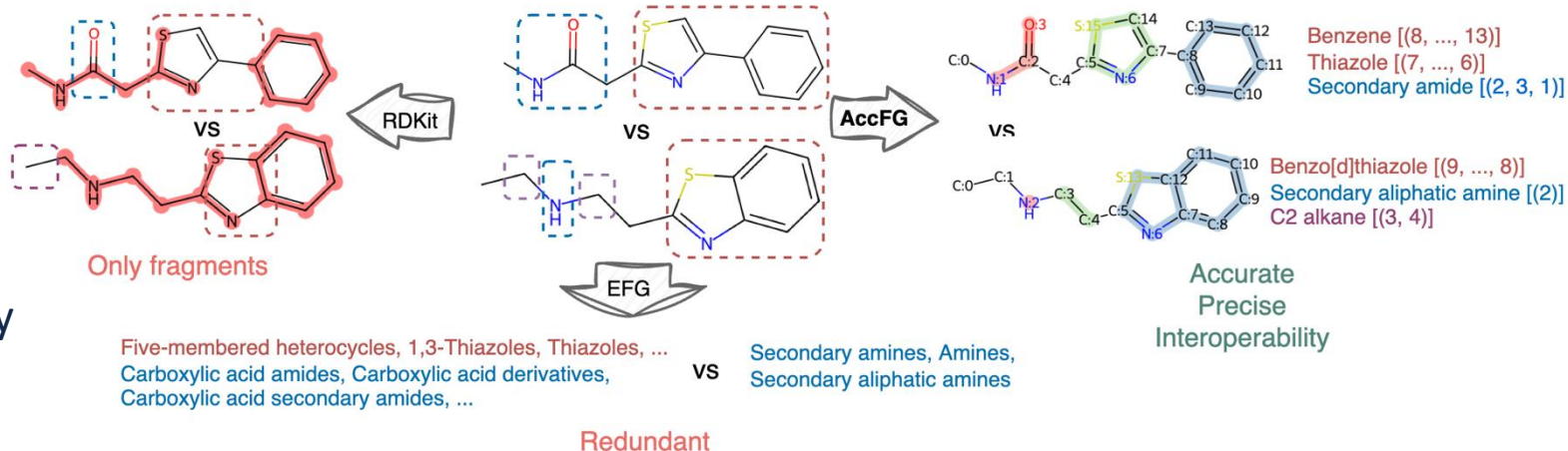
(a) Functional Group Extraction

Carboxylic acid derivatives
Carboxylic acid amide
Carboxylic acid primary amide
Aromatic compounds
Heterocyclic compounds
Aromatic heterocyclic compounds
...
(Total of 18 entries)

Redundant



(b) Molecule Comparison



Philosophy behind the AccFG

1. Avoids strict SMARTS definitions

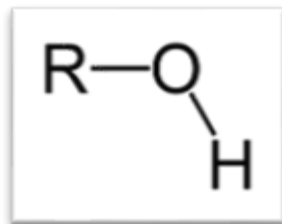
- **Hydroxy group**

- Ours: [OX2H]

- Others:

[OX2H][CX4;!\$(C([OX2H])[O,S,#7,#15])]

- [CX4]: Tetra-valent carbon atom connected to the hydroxyl group
- !\$(C([OX2H])[O,S,#7,#15]): Specifies that the carbon **must not** be connected to O,S,N,P.



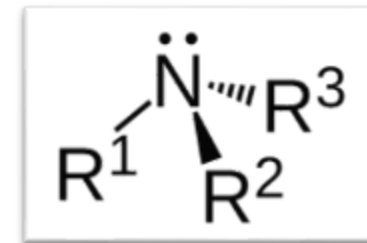
- **Amine**

- Ours: [NX3+0]

- Others:

[NX3+0;!\$([N]~[!#6]);!\$([N]*~[#7,#8,#15,#16])]

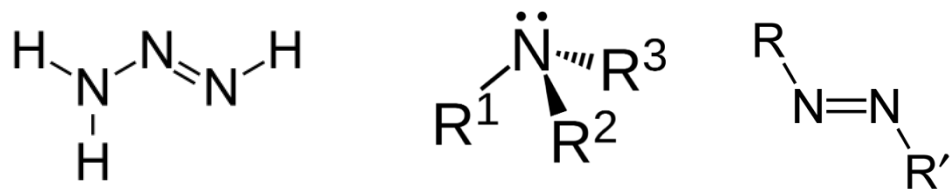
- [NX3+0]: Neutral nitrogen with three connections
- !\$([N]~[!#6]): Nitrogen must not bond to non-carbon atoms
- ...



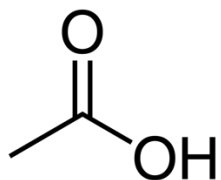
Philosophy behind the AccFG

1. Avoids strict SMARTS definitions
2. Prioritizes specific & informative groups

- E.g., “triazene” > “amine” & “azo”



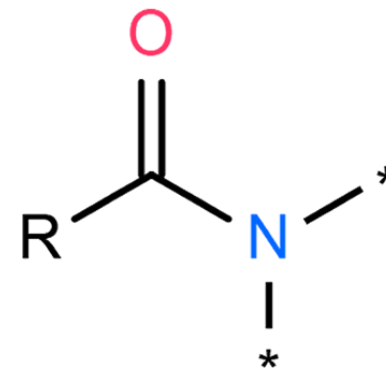
- E.g., “carboxyl” > “ketone” & “hydroxy”



The semantic knowledge of “carboxyl” is more suitable to describe this molecule than “ketone” & “hydroxy”

Loose FG definitions cause problems:

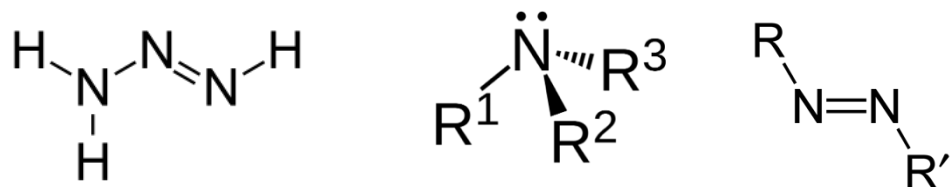
An example of mismatch on **amide** group



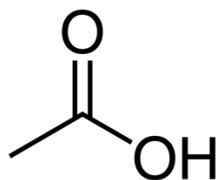
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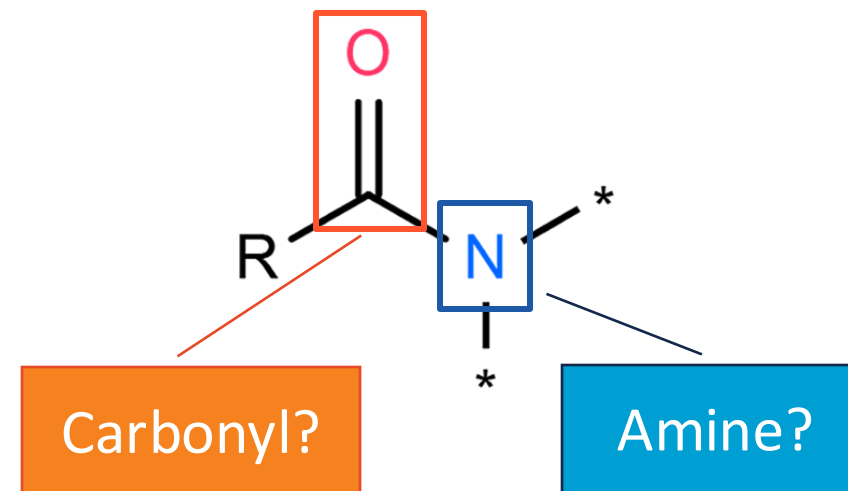
- E.g., “carboxyl” > “carbonyl” & “hydroxy”



The semantic knowledge of “carboxyl” is more suitable to describe this molecule than “ketone” & “hydroxy”

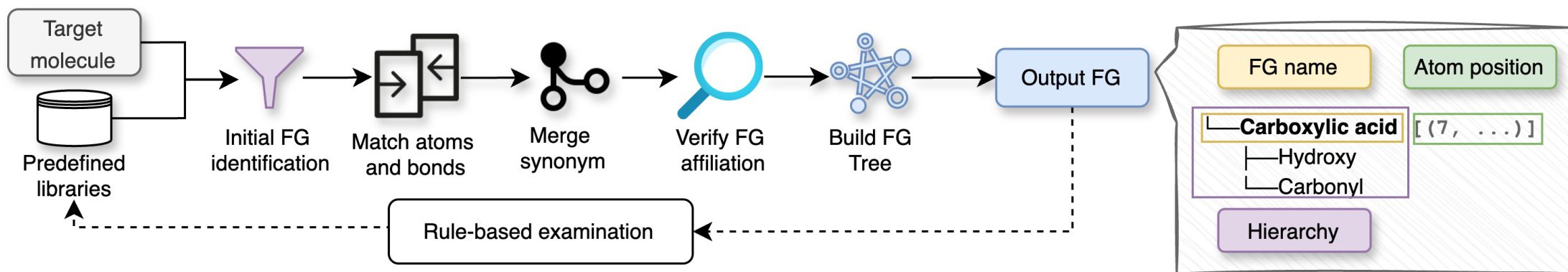
Loose FG definitions cause problems:

An example of mismatch on **amide** group

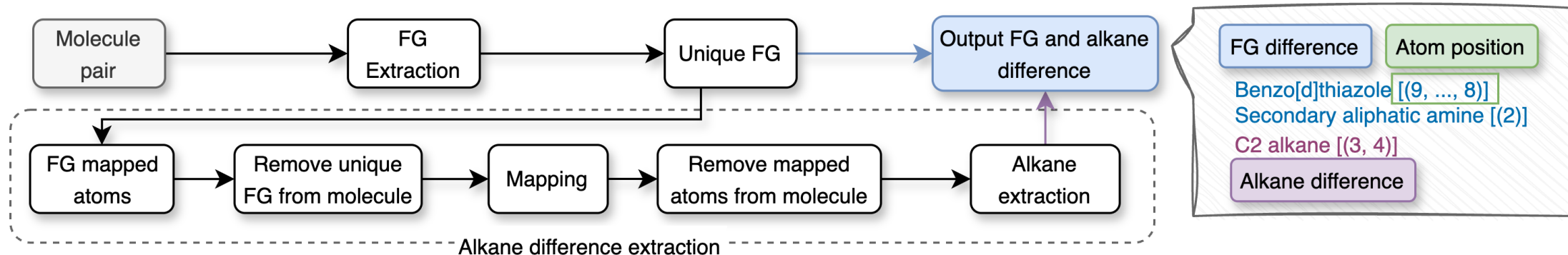


Development of AccFG

(a) Workflow of Functional Group Extraction



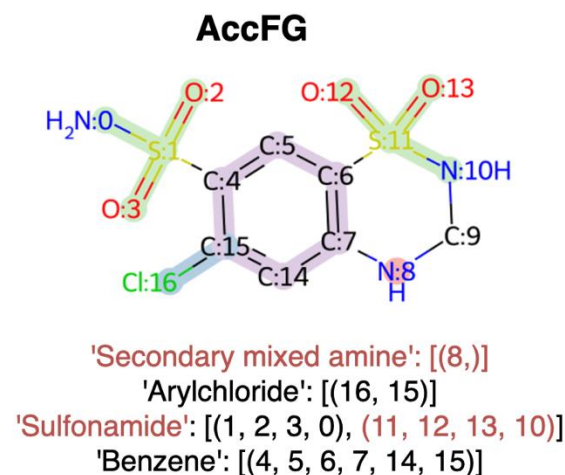
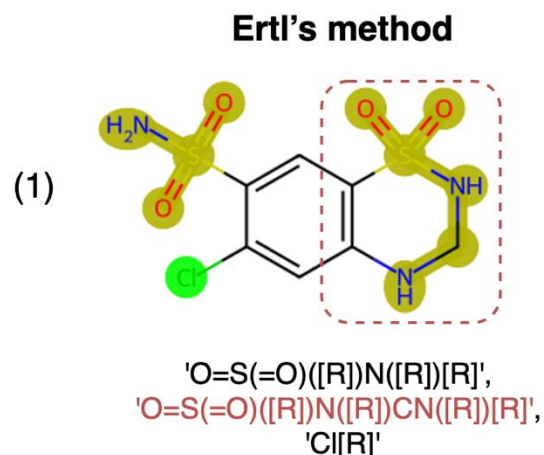
(b) Workflow of Molecule Comparison



The final libraries include 321 common functional groups and 210 heterocycles

Examples of functional group extraction result of AccFG compared with Ertl's method and EFG

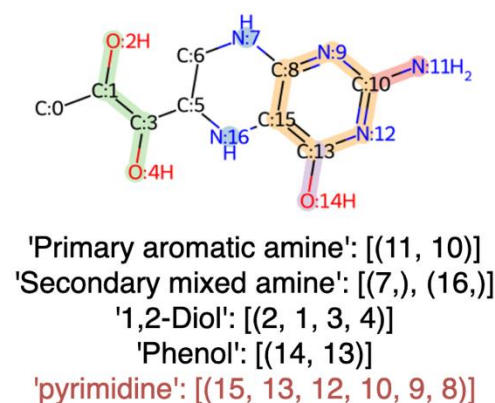
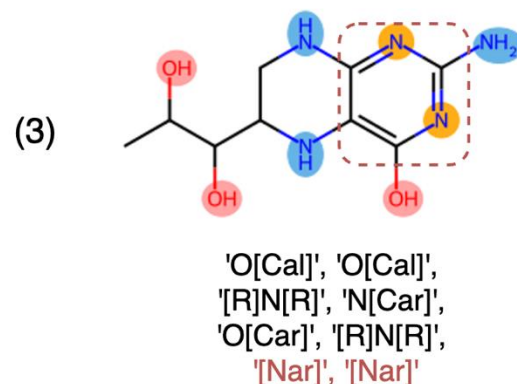
FG identification



EFG

Sulfonic acid derivatives
Sulfonamides
Heterocyclic compounds
Pnictogens (nitrogen group)
Six-membered heterocycles (LS)
Six-membered heterocycles with three heteroatoms (LS)
Halogen derivatives (alkyl, alkenyl, aryl)
Aryl chlorides
...
(Total of 15 entries)

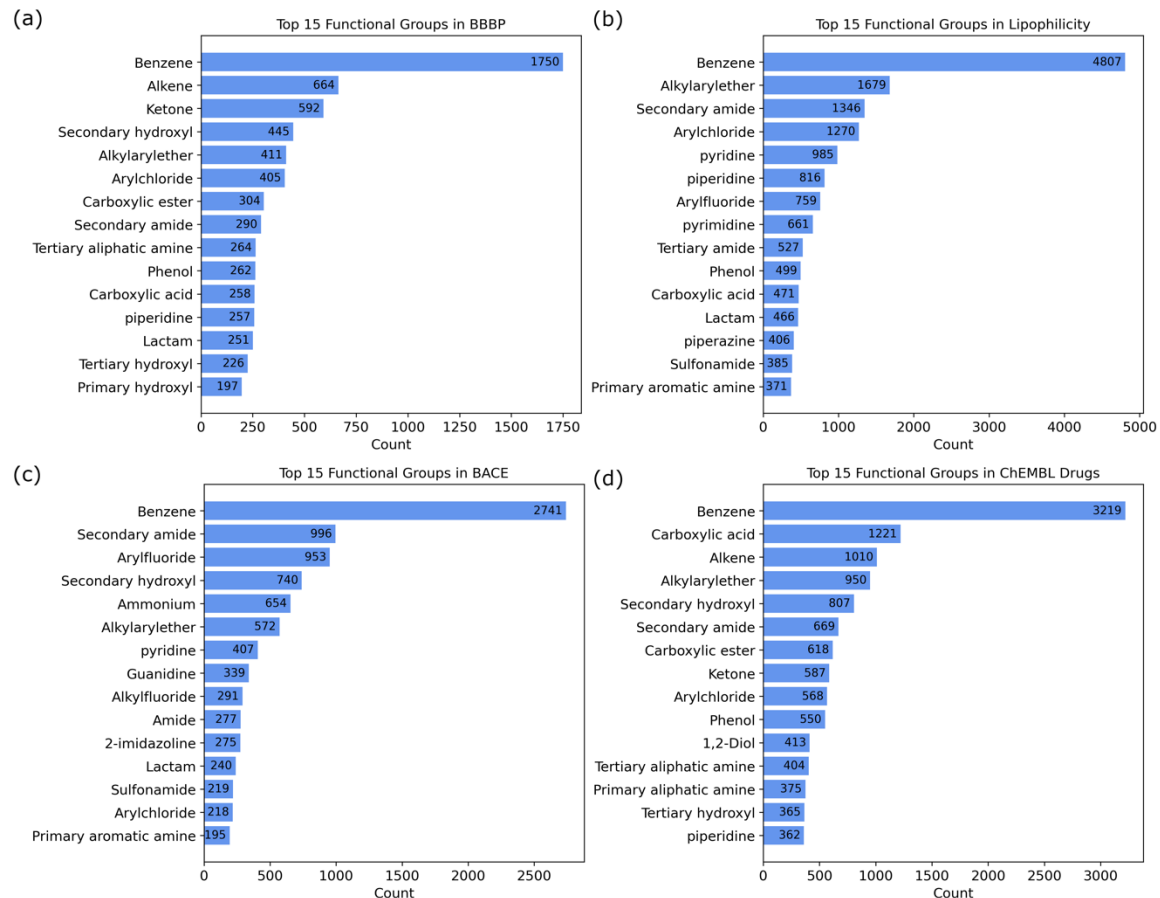
Heterocycle identification



Six-membered heterocycles (LS)
Six-membered heterocycles with two heteroatoms (LS)
Aromatic six-membered heterocycles with two heteroatoms (LS)
1,2-Diamines
Aminophenols
...
(Total of 25 entries)

Beyond case study: Application on molecule datasets

AccFG on 4 different datasets



Aligning well with Ertl's method on ChEMBL drugs

Rank	AccFG			Ertl's method		
	Functional Group	Count	Percentage	pseudo-SMILES	Count	Percentage
1	Benzene	1960	15.24	[Nar]	1594	14.79
2	Alkene	663	5.16	O[Cal]	1172	10.87
3	Alkylarylether	569	4.42	[R]O[R]	924	8.57
4	Secondary hydroxyl	493	3.83	[R]N([R])[R]	738	6.85
5	Carboxylic acid	447	3.48	F[R]	722	6.70
6	Ketone	440	3.42	O=C([R])N([R])[R]	611	5.67
7	Carboxylic ester	424	3.30	Cl[R]	473	4.39
8	Secondary amide	415	3.23	O=C([R])O[R]	407	3.78
9	Arylchloride	379	2.95	O=C(O)[R]	383	3.55
10	Phenol	334	2.60	O[Car]	334	3.10
11	Arylfluoride	236	1.84	[R]N[R]	329	3.05
12	Dialkylether	226	1.76	O=C([R])[R]	263	2.44
13	Pyridine	214	1.66	C=C	207	1.92
14	Piperidine	214	1.66	O=[Car]	195	1.81
15	Tertiary aliphatic amine	198	1.54	N[Car]	175	1.62
16	Tertiary hydroxyl	198	1.54	N[Cal]	173	1.60
17	Primary hydroxyl	197	1.53	[R]OCO[R]	131	1.22
18	Lactam	178	1.38	[Sar]	118	1.09
19	1,2-Diol	174	1.35	O=S(=O)([R])N([R])[R]	99	0.92
20	Primary aliphatic amine	173	1.35	[R]I	98	0.91

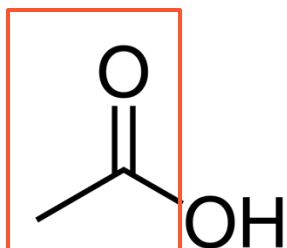
Philosophy behind the AccFG

1. Avoids strict SMARTS definitions
2. Prioritizes specific & informative groups
3. Semantic alignment of functional groups for molecule structure comparison

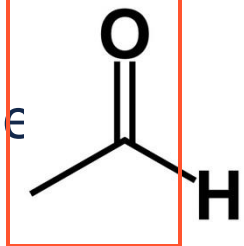
Direct alignment:

The difference is -OH .

Acetic acid

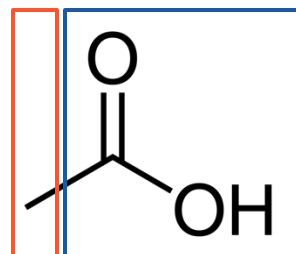


Acetic aldehyde



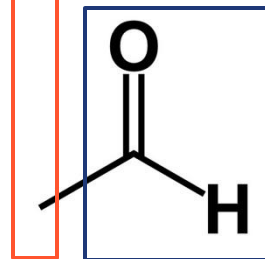
Semantic Alignment of Functional groups:

-COOH vs -CHO



Carboxyl

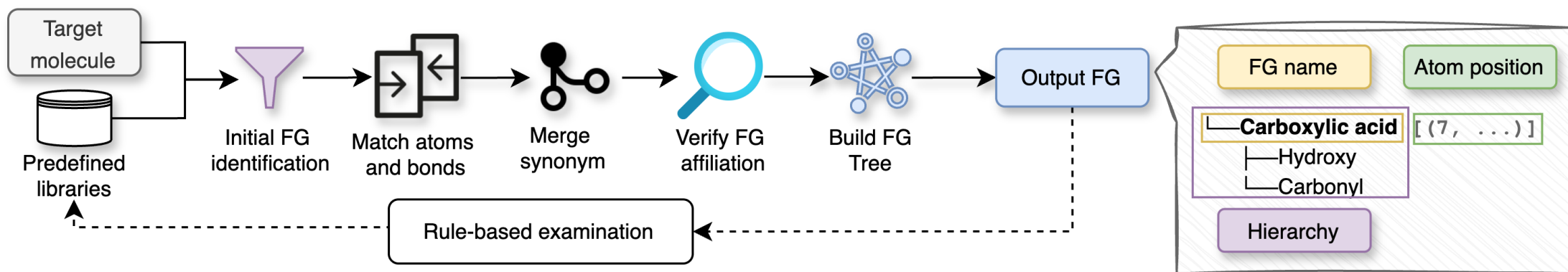
V.S.



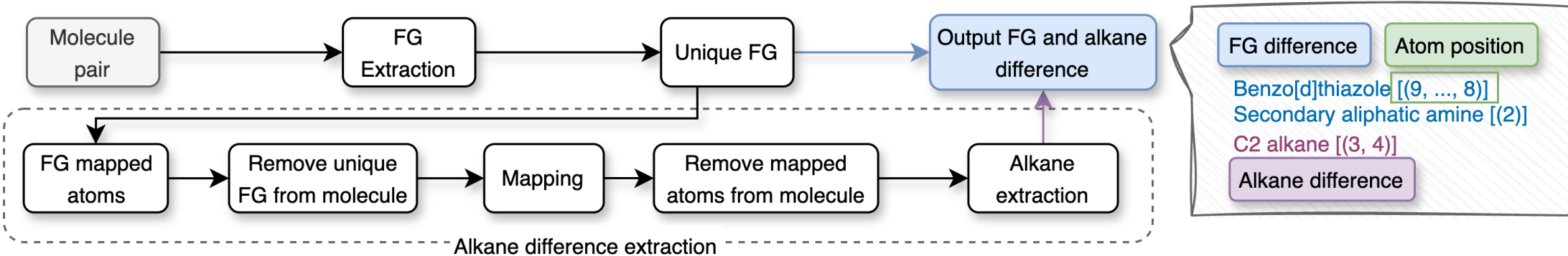
Aldehyde

Development of AccFG

(a) Workflow of Functional Group Extraction



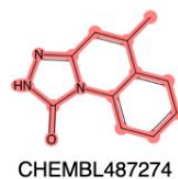
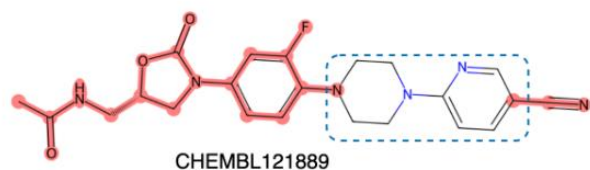
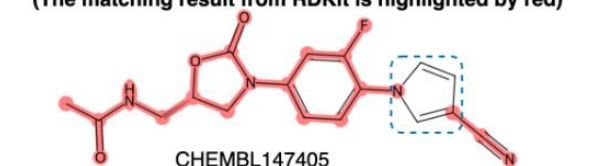
(b) Workflow of Molecule Comparison



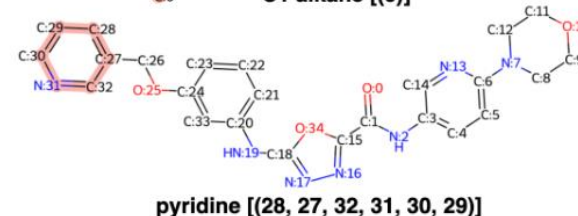
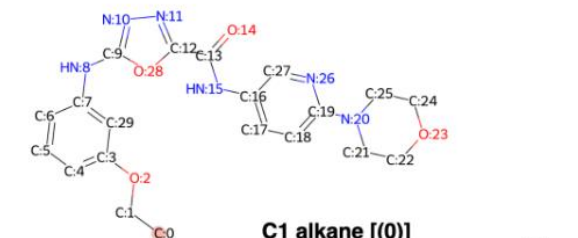
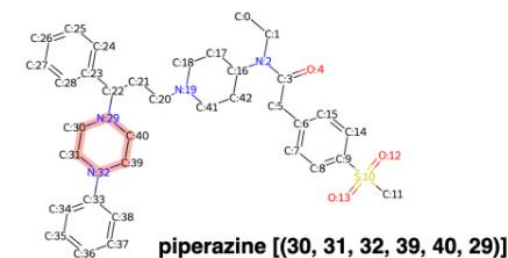
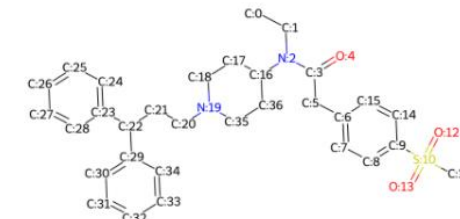
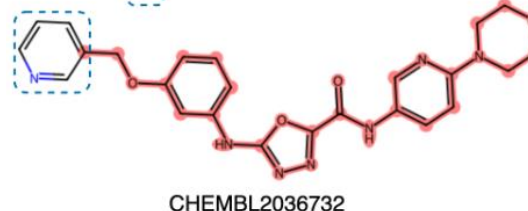
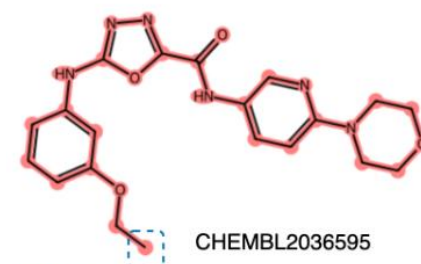
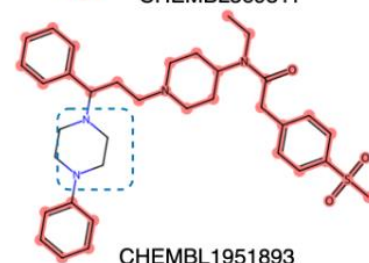
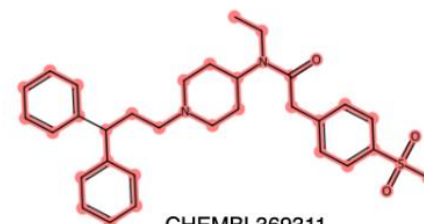
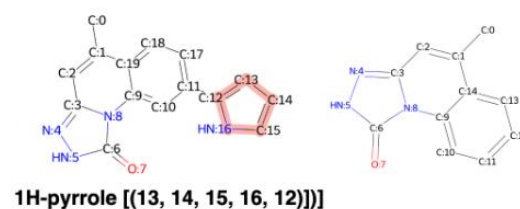
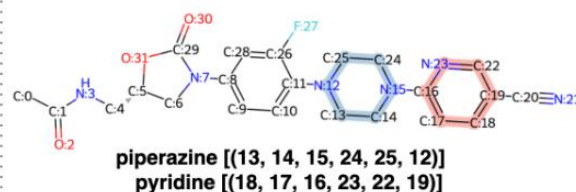
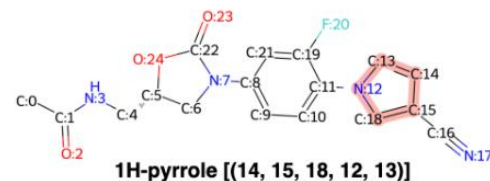
Molecule comparison

- FG difference
- Alkane difference

RDKit MCES structure comparison
(The matching result from RDKit is highlighted by red)



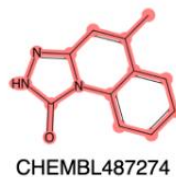
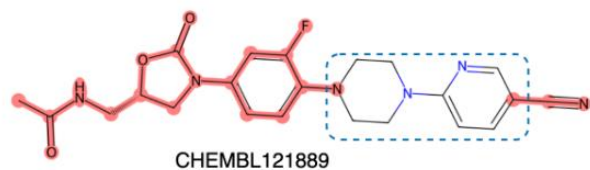
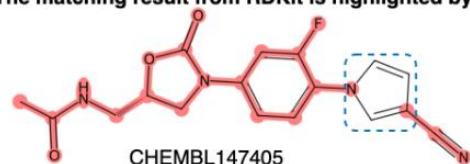
AccFG structure comparison



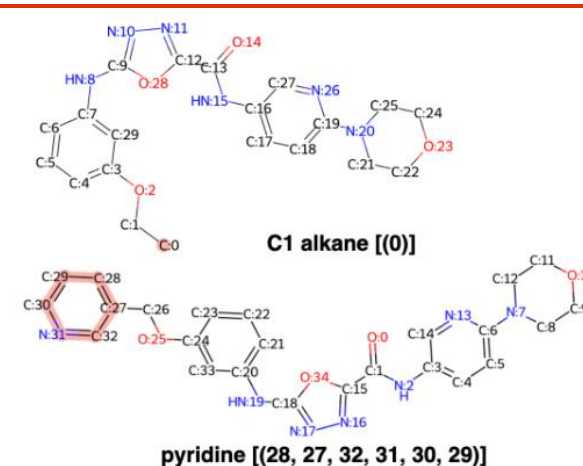
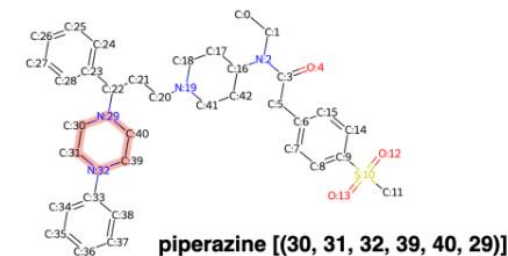
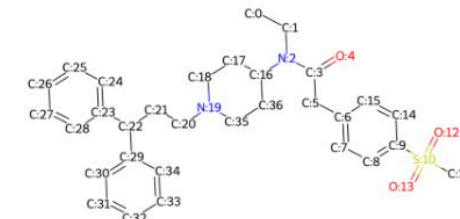
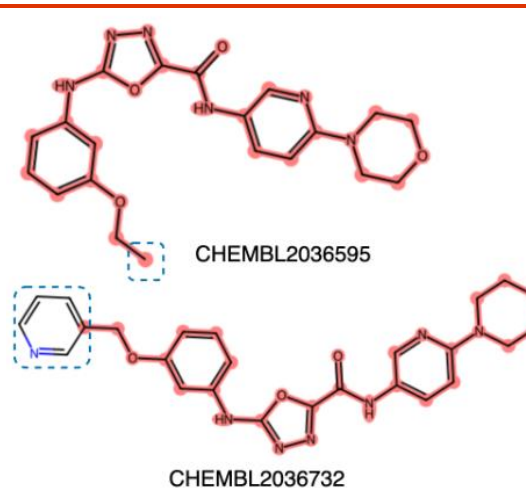
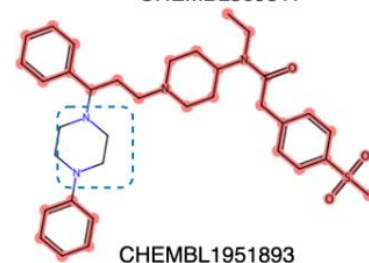
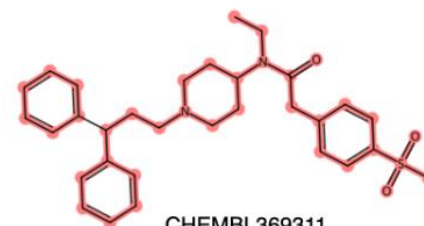
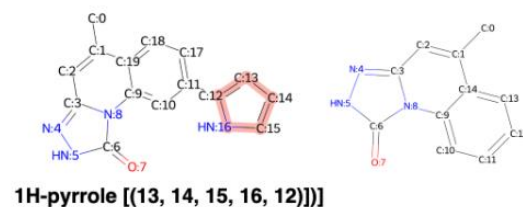
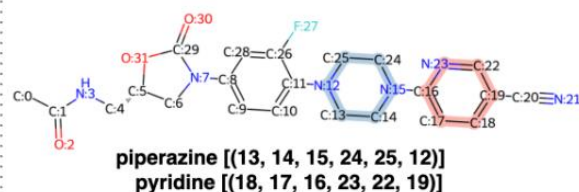
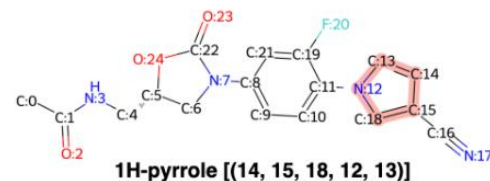
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AccFG structure comparison



AccFG Installation

 AccFG

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AccFG

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Quickstart

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Tutorials

Case Study

Function Reference

accfg.main

accfg.compare

accfg.draw

License

AccFG

AccFG is a tool for Accurate Functional Group Extraction and Molecular Structure Comparison

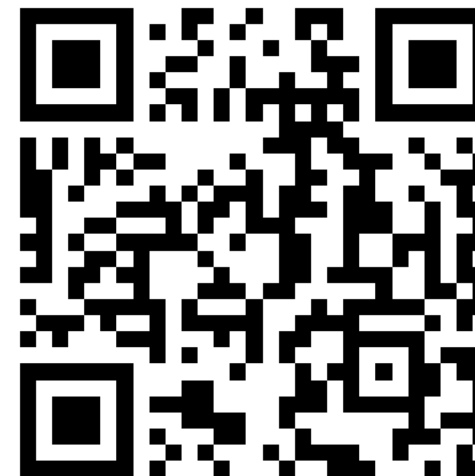
Features

- Extract functional groups from a molecule with **atom number**.
- Automatically build **hierarchical** functional groups.
- A "**Lite**" version for a concise list of functional groups.
- **Compare two molecules** at functional group level.
- Easy to **Customize FG list** and no complex SMARTS constraints.

Quick start

Installation

```
pip install accfg
```



See documentation at
<https://xuanliugit.github.io/AccFG/>

Install AccFG by `pip`

Using AccFG

- AccFG provides easy-to-use **functions** and **visualization tools**

```
from rdkit import Chem
from rdkit.Chem import Draw

from accfg import (
    AccFG,
    draw_mol_with_fgs,
    molimg,
    img_grid,
    compare_mols,
    draw_compare_mols,
    draw_RascalMCES,
    print_fg_tree,
)

afg = AccFG(print_load_info=True)
```

```
smi = "CN(C)/N=N/C1=C(NC=N1)C(=O)N"
fgs, fg_graph = afg.run(smi, show_atoms=True, show_graph=True, canonical=False)

print("Functional groups:\n", fgs)
print("-----")
print_fg_tree(fg_graph, fgs.keys(), show_atom_idx=True)
```

```
Functional groups:
{'Primary amide': [(10, 11, 12)], 'Triazene': [(1, 3, 4)], 'imidazole': [(5, 6, 7, 8, 9)]}
-----
|—Primary amide: ((10, 11, 12),)
|   |—Amine: ((1,), (12,))
|   |   |—Carbonyl: ((10, 11),)
|—Triazene: ((1, 3, 4),)
|   |—Amine: ((1,), (12,))
|   |   |—Azo: ((3, 4),)
|—imidazole: ((5, 6, 7, 8, 9),)
|   |—Hetero N basic H: ((7,),)
|   |   |—Hetero N nonbasic: ((9,),)
```

Using AccFG

- AccFG provides easy-to-use **functions** and **visualization tools**

```
from rdkit import Chem
from rdkit.Chem import Draw
```

```
from accfg import (
    AccFG,
    draw_mol_with_fgs,
    molimg,
    img_grid,
    compare_mols,
    draw_compare_mols,
    draw_RascalMCES,
    print_fg_tree,
)
```

```
afg = AccFG(print_load_info=True)
```

```
smi = "CN(C)/N=N/C1=C(NC=N1)C(=O)N"
fgs, fg_graph = afg.run(smi, show_at
```

```
print
print
print
```

Visualising FG Highlights

```
mol_fg_img = molimg(
    draw_mol_with_fgs(
        smi,
        canonical=False,
        alpha=1,
    )
)
mol_fg_img
```

Function

{'Prima

|—Prima

|—Am

|—Ca

|—Triaz

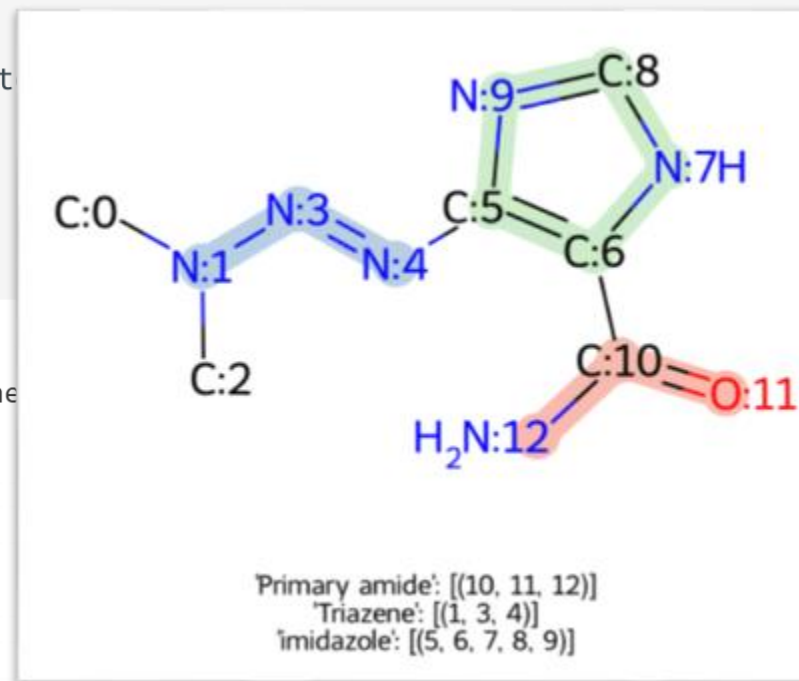
|—Amine: ((1,), (12,))

|—Azo: ((3, 4),)

|—imidazole: ((5, 6, 7, 8, 9),)

|—Hetero N basic H: ((7,),)

|—Hetero N nonbasic: ((9,),)



Using AccFG is easy

- AccFG provides easy-to-use **functions** and **visualization tools**

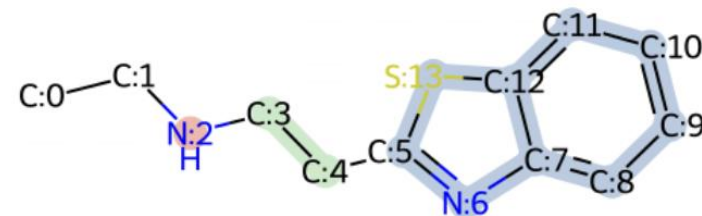
Molecule Structure Comparison

```
smi_1, smi_2 = (  
    "CNC(=O)Cc1nc(-c2ccccc2)cs1",  
    "CCNCCc1nc2ccccc2s1",  
)  
diff = compare_mols(smi_1, smi_2)  
diff
```

```
img = img_grid(draw_compare_mols(smi_1, smi_2), num_columns=2)  
img
```



'Secondary amide': [(2, 3, 1)]
'Benzene': [(8, 9, 10, 11, 12, 13)]
'thiazole': [(7, 14, 15, 5, 6)]



'Secondary aliphatic amine': [(2,)]
'benzo[d]thiazole': [(9, 10, 11, 12, 13, 5, 6, 7, 8)]
'C2 alkane': [(3, 4)]

User-defined Functional group identification

- Drug pattern mining

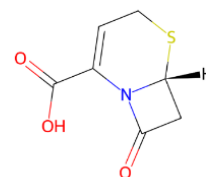
Adding new FG is simple

```
my_fgs_dict = {  
    "Cephem": "O=C(O)C1=CCS[C@@H]2CC(=O)N12",  
    "Thioguanine": "Nc1nc(=S)c2[nH]cnc2[nH]1",  
}  
my_afg = AccFG(user_defined_fgs=my_fgs_dict, print_load_info=True)
```

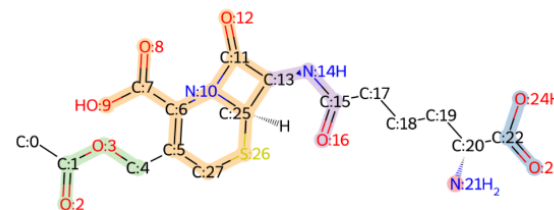
```
cephalosporin_C = "CC(=O)OCC1=C(N2[C@@H]([C@@H](C2=O)NC(=O)CCC[C@H](C(=O)O)N)  
fgs, fg_graph = my_afg.run(cephalosporin_C, show_atoms=True, show_graph=True)  
print_fg_tree(fg_graph, fgs.keys(), show_atom_idx=True)
```

```
molimg(  
    draw_mol_with_fgs(  
        cephalosporin_C,  
        afg=my_afg,  
        img_size=(900, 900),  
    )  
)
```

(a)



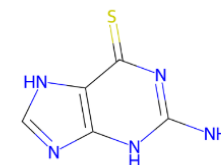
Cephem:
O=C(O)C1=CCS[C@@H]2CC(=O)N12



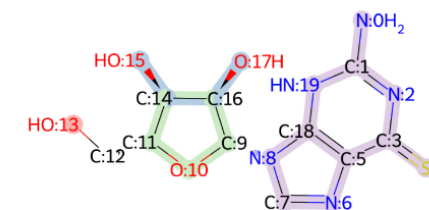
Cephalosporin C

- Primary aliphatic amine [(21)]
- Carboxylic acid [(7, ...), (22, ...)]
 - Hydroxy
 - Carbonyl
- Carboxylic ester [(1, 2, 3, 4)]
 - Carbonyl
 - ...
- Secondary amide [(15, 16, 14, 13)]
 - Carbonyl
- Cephem [(8, 7, 9, ..., 10)]
 - Alkene
 - Dialkylthioether
 - azetidin-2-one
 - ...

(b)



Thioguanine:
Nc1nc(=S)c2[nH]cnc2[nH]1



6-Thioguanosine

- Primary hydroxyl [(13)]
- 1,2-Diol: [(15, 14, 16, 17)]
 - Hydroxy
 - ...
- tetrahydrofuran [(14, 16, 9, 10, 11)]
 - Dialkylether
 - ...
- Thioguanine [(0, 1, 2, ..., 19)]
 - Primary aromatic amine
 - Thioarene
 - purine
 - ...

For developer (Instruction and Function reference)

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Installation

AccFG can be installed from PyPI or directly from the source repository. The recommended path is via `pip`, but advanced users can bootstrap a local development copy.

1. Install from PyPI (recommended)

```
pip install accfg
```

This installs the latest published release together with its Python dependencies. AccFG targets Python 3.10; create a dedicated virtual environment if you need to isolate the package.

2. Install from the GitHub repository

Use this route when you need the latest commits or intend to contribute.

```
git clone https://github.com/xuanliugit/AccFG.git
cd AccFG

conda create --name accfg python=3.10
conda activate accfg
pip install -r requirements.txt # or: pip install -e .
```

The editable install (`pip install -e .`) keeps the package linked to your working tree so changes are picked up immediately.

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Edit Functional Group Lists

AccFG loads functional group definitions from CSV files packaged with the library. You can extend or customise these dictionaries by editing the CSV files directly or by supplying user-defined SMARTS at runtime.

1. Built-in CSV dictionaries

The default dictionaries live under `accfg/`:

- `accfg/fgs_common.csv`: General functional groups.
- `accfg/fgs_heterocycle.csv`: Heterocycles and related ring systems.

Both files follow a custom CSV format tailored for `AccFG.csv_to_dict()`:

- Lines beginning with `%` are treated as comments and ignored.
- In non-lite mode, rows whose `Functional Group` column begins with `#` are considered active entries (the `#` is stripped during import).
- Each row must define two columns: `Functional Group` (label) and `SMARTS Pattern` (SMARTS string describing the substructure).

Before editing, create a backup copy so you can revert if a new entry causes parsing errors.

2. Adding new entries

Using AccFG in your work

AI for Accelerated Materials Discovery (AI4Mat) Workshop at NeurIPS 2025

Foundation Models Enabling Multi-Scale Battery Materials Discovery: From Molecules To Devices

*Vidushi Sharma, Maxwell J Giammona, Andy Tek, Murtaza Zohair,
Nathaniel H. Park, Tim Erdmann, Eduardo Soares, Linda Sundberg,
Khanh Nguyen, Young-Hye Na, Emilio Vital Brazil*

 Published: 29 Oct 2025, Last Modified: 29 Oct 2025  AI4Mat-NeurIPS-2025 Poster  Everyone
 Revisions  BibTeX  CC BY 4.0

Keywords: Foundation Models, Material Discovery, Batteries, Devices, Prediction, Benchmarking, Empirical Datasets, Scarce Data, Electrolytes

Abstract:

Recent years have seen fast emergence and adoption of chemical foundation models in computational material science for property prediction and generation tasks that are focused mostly on small molecules or crystals. Despite these paradigm shifts, integration of newly discovered materials in real world devices continues to be a challenge due to design problems. New candidate material must be optimized to achieve compatibility with other components in the system and deliver the target performance. Chemical foundation model benchmarks must evaluate their scope in predicting macro scale outcomes that are the result of chemical interactions in multi-variate design space. This study evaluates performance of chemical foundation models that are pre-trained primarily with SMILES of small molecules, in extrapolating learning from molecules to material design challenges across multiple length scale in batteries. Ten prediction models are trained covering molecular properties, formulations performance, and battery device measurement. Material representations from several foundation models are compared and their performance is benchmarked against conventional molecular representations such as Morgan Fingerprints. The study further examines their capacity to generalize to out-of-distribution cases by quantifying prediction errors for novel material designs that differ substantially from the training data. Finally, interpretability of the trained predictors is assessed by correlating actual outcomes and predictions to the chemical moieties in the datasets, with the aim of enabling researchers to interpret design rules in chemical space where model has high confidence.

NeurIPS 2025 (Datasets & Benchmark Track)

FGBench: A Dataset and Benchmark for Molecular Property Reasoning at Functional Group-Level in Large Language Models

Xuan Liu, Siru Ouyang, Xianrui Zhong, Jiawei Han, Huimin Zhao

 Published: 18 Sept 2025, Last Modified: 29 Oct 2025
 NeurIPS 2025 Datasets and Benchmarks Track poster  Everyone  Revisions  BibTeX
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Keywords: Dataset, Benchmark, Molecules, Functional groups, Reasoning

TL;DR: New dataset and benchmark evaluate LLMs on molecular property reasoning at a fine-grained functional group level

Abstract:

Large language models (LLMs) have gained significant attention in chemistry. However, most existing datasets center on molecular-level property prediction and overlook the role of fine-grained functional group (FG) information. Incorporating FG-level data can provide valuable prior knowledge that links molecular structures with textual descriptions, which can be used to build more interpretable, structure-aware LLMs for reasoning on molecule-related tasks. Moreover, LLMs can learn from such fine-grained information to uncover hidden relationships between specific functional groups and molecular properties, thereby advancing molecular design and drug discovery. Here, we introduce FGBench, a dataset comprising 625K molecular property reasoning problems with functional group information. Functional groups are precisely annotated and localized within the molecule, which ensures the dataset's interoperability thereby facilitating further multimodal applications. FGBench includes both regression and classification tasks on 245 different functional groups across three categories for molecular property reasoning: (1) single functional group impacts, (2) multiple functional group interactions, and (3) direct molecular comparisons. In the benchmark of state-of-the-art LLMs on 7K curated data, the results indicate that current LLMs struggle with FG-level property reasoning, highlighting the need to enhance reasoning capabilities in LLMs for chemistry tasks. We anticipate that the methodology employed in FGBench to construct datasets with functional group-level information will serve as a foundational framework for generating new question-answer pairs, enabling LLMs to better understand fine-grained molecular structure-property relationships. The dataset and evaluation code are available at this [link](https://github.com/xuanliugit/FGBench).

Summary

- **High Accuracy & Precision:**

AccFG delivers superior results in both **functional group extraction** and **molecular structure comparison**.

- **Efficient & Extensible Algorithm:**

Easily integrates new functional groups while maintaining **robust performance**.

- **Interpretability:**

Outputs are concise and directly link **molecular fragments to natural language**, enhancing human understanding.

Acknowledgement

- **PhD Advisor:** Prof. Huimin Zhao
- **Co-authors:** Sarathkrishna Swaminathan, Dmitry Zubarev, Brandi Ransom, Nathaniel Park, Kristin Schmidt, Huimin Zhao
- **IBM-IL Discovery Accelerator Institute (IIDAI)**
- **Molecule Maker Lab Institute (MMLI)**

Thanks for listening!

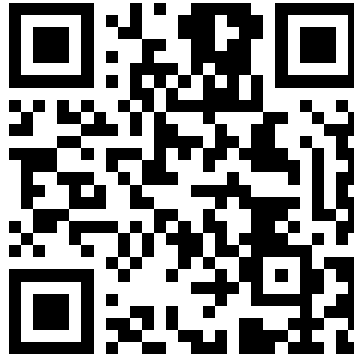
About me

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Research interests: AI4Science, LLMs

Open to work (intern or full-time)



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Paper:
JCIM 65.16 (2025): 8593-8602.



GitHub:
[xuanliugit/AccFG](https://github.com/xuanliugit/AccFG)
(Slides available soon)

My Next Talk:

FGBench: A Dataset and Benchmark for Molecular Property Reasoning at the Functional Group Level in Large Language Models (NeurIPS'25 DB track)



Thursday, November 6, 2025 | 11:42 AM – 11:54 AM



Session: Machine Learning for Soft and Hard Materials II: Hard Matter and Methods



Location: Hynes Convention Center, Room 210